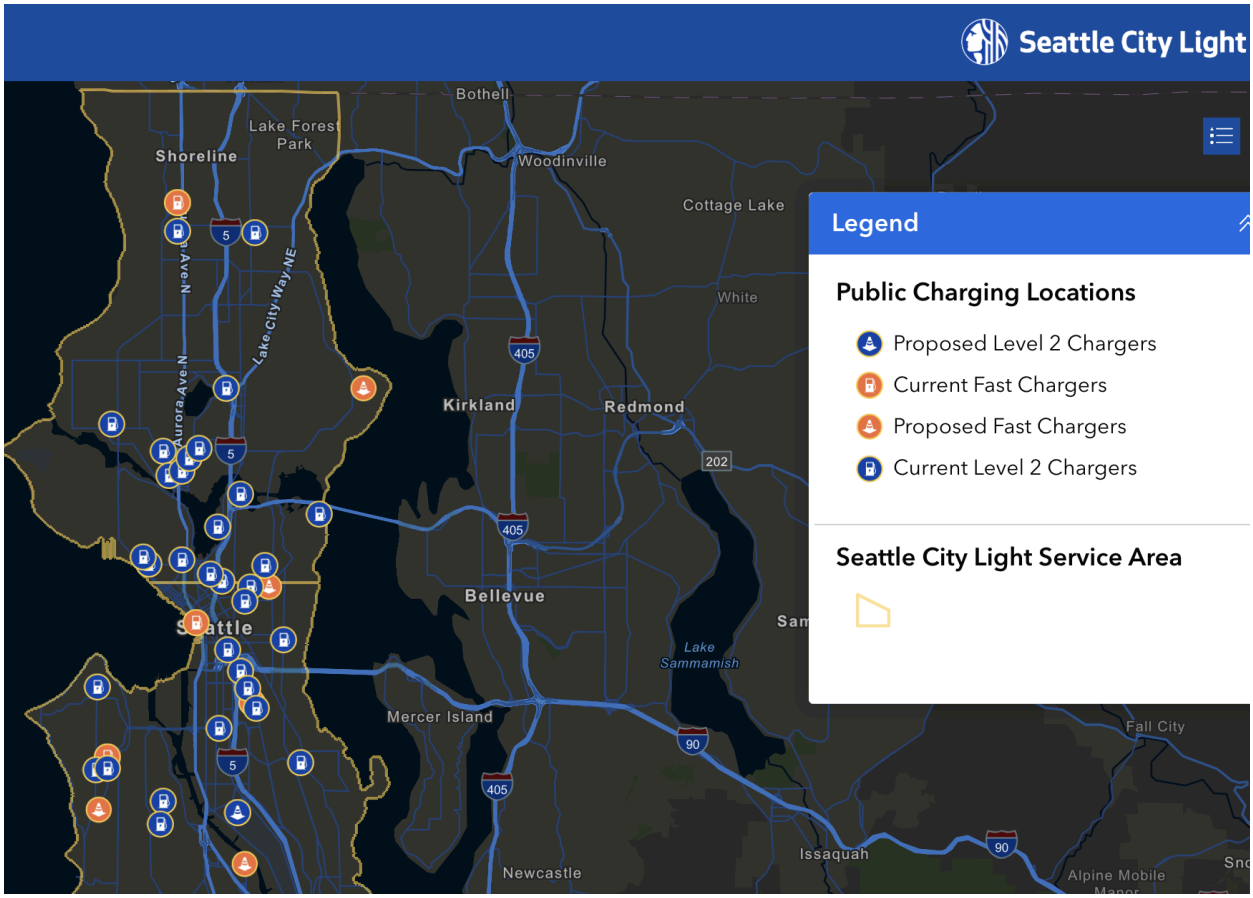
UIUC GGIS 480 - Spring 2026 - Ryan Rickerts

Electric Vehicle Charger Site Selection

SEATTLE METRO AREA SPATIAL OPTIMIZATION



Introduction

To increase adoption of electric vehicles (EVs) and reduce society's usage of fossil fuels for solving our global warming problem, we must increase the number of publicly available charging stations. This reduces "range anxiety" which prevents people from committing to fully electric vehicles.

My project presumes a public electric utility, Seattle City Light or Puget Sound Energy, has the budget to place 15 additional EV chargers on public roads and I have been tasked with using my spatial data analysis skills to help them select the optimum locations.

Problem Statement

My hypothetical employer, the regional electricity provider, has tasked me with finding "charging deserts" where people in their EVs have been historically stranded out of battery charge, indicating a demand node or "charging desert" that could be addressed using the budget for new charging infrastructure.

Posing this as a P-center problem, where distance must be minimized along the road network, we can find the optimum locations to meet the unweighted demand (i.e. minimize distance for each incident).

For reporting to my superiors, I have prepared a series of plots overlaying existing charging infrastructure, major roads in the network, and AAA-reported outage locations in the larger Seattle metro area. These images have been included in this report, also.

Data Sources & Pre-processing Steps

The following data sources were utilized to enable this spatial analysis using Jupyter notebook on the CyberGISX platform.

AAA Out of Charge Incidents

This interest of mine goes back to 2023 when I thought about using the Google Maps API and some yet unknown ML technique to determine the optimum charger locations, perhaps train a model that could be utilized for dynamic outputs as more funding for EV chargers emerged from federal or state initiatives.

At that time, AAA gave an online presentation I attended talking about using their dataset for service calls nationally in the U.S. to solve the same problem. This slide is from their sales pitch. I contacted their company representative and purchased the data for Washington State for 1 year (Nov. '22 to Nov. '23) and this cost me $200.

At that price for an unsponsored academic project, I have decided to re-use the .csv datafile they sent me. It contains the latitude and longitude of the vehicle pickup, make/model of the vehicle, the cause for the service call (e.g. flat tire), etc. I had to filter this data for only out-of-charge incidents which reduced the number of interesting points in the Seattle Metro area to 39 events.

This is perhaps statistically insignificant but adequate for a proof-of-concept approach. With a greater budget, a larger, multi-year dataset could be purchased from AAA for better pattern analysis.

Washington Road Network

The road network for the entirety of Washington State was downloaded for free from the Geofabrik GmbH website which extracts and compiles OpenStreetMap project data. You can drill into sub-regions by country and further into the sub-region of interest. Washington is the smallest resolution offered and overlaps perfectly with my AAA dataset. However, I had difficulty loading this entire .osm.pbf file into the kernel memory in the CyberGISX environment, a limitation I will discuss in the Pre-processing section below.

Existing Charger Infrastructure

The last and perhaps most critical data source came from the U.S. Department of Energy Alternative Fuels Data Center Data Downloads page. This, like the road network data, is filterable by state of interest with public & functioning infrastructure only. It also allows for a historical query, so I selected Dec. 31, 2023 to match the AAA dataset. The data format is provided in a GeoJSON with lat/lon values for the stations.

Pre-processing Steps

I encountered some unexpected constraints with my project plan due to memory limits in the CyberGISX environment itself. Multiple attempts and approaches to getting the Washington road network into OSM failed and caused a kernel restart. Even filtering for main arteries like this:

```
metro_roads = osm.get_network(network_type="driving")
highway_types = ["motorway", "motorway_link", "primary", "primary_link",
"secondary"]
metro_roads = metro_roads[metro_roads["highway"].isin(highway_types)].copy()
```

and garbage collecting the memory did not solve the issue.

Thanks in part to my background in Computer Science and daily work in software engineering in a cloud computing environment, I was able to tackle this problem in a different way. While I considered spinning up custom infrastructure on AWS Sagemaker with higher powered GPU processing power (something I have done in previous grad projects at UIUC), this assignment specified using exclusively the CyberGISX platform.

Therefore, I installed the Osmosis library, a command line Java application for processing OSM data (sponsored by the OpenStreetMap project itself), in the terminal. This also required installing Java version 17, all of which was possible without sudo privileges. I do not believe the installation was retained, but I was able to get my output file with the lat/lon bounds of the Seattle Metro area and still solve an interesting set of questions with pyrosm, networkx, pulp, and other python libraries we learned about in this course.

```
osmosis \
   --read-pbf washington-latest.osm.pbf \
   --bounding-box left=-122.8 bottom=47.1 right=-122.0 top=48.1 \
   --tf accept-ways
highway=motorway,motorway_link,primary,primary_link,secondary \
   --used-node \
   --write-pbf seattle_corridor.osm.pbf
```

This still reduces the road network to primary and secondary motorways which came back to bite me later in the project which is discussed next.

Charging Location Proximity Analysis

The first key step in identifying charging deserts is to find the network distance from the outage incidents to the nearest charger. This is where the `networkx` or `osmnx` libraries would really shine, such as `nx.shortest_path_length(graph, orig, dest, weight='length')` or:

```
# Find nearest node for each incident and station
orig_nodes = ox.nearest_nodes(G, aaa_ooc_incidents.geometry.x,
aaa_ooc_incidents.geometry.y)
dest_nodes = ox.nearest_nodes(G, existing_chargers.geometry.x,
existing_chargers.geometry.y)
```

However, due to the pre-processing of the network described earlier, the outage incidents were not sited on the network (imagine pulling off the main highway to have your car die peacefully on a back road).

Therefore, I reverted to using straightline distance with GeoPandas dataframes. We can make 1km and 3km bubbles around the charging stations and determine how many outage incidents were exterior to these. The outputs of this analysis were as follows (suitable to report to management):

```
Average distance to nearest charger: 1.11 km
Max distance (Worst Case): 3.83 km

Incidents in 'Deserts' (>1km from charger): 16
Incidents in 'Critical Deserts' (>3km from charger): 5
```

This provides the impetus to serve our candidate sites outside of a standard service area using the P-Center Problem approach.

First, we construct a grid of candidate charger locations using the intersection of two or more roads in the network (aka. nodes) which helps maximize service from multiple directions. We designate one node for every 2km within the Seattle Metro corridor. Then, we make our `scipy.spatial.distance` matrix::

```
# Calculate Distance Matrix (d_ij) between 39 incidents and ~800 grid points
 dist_matrix = cdist(demand_coords, grid_points)
```

The P-Center Problem is solved given the usual constraints (similar to Module 7 assignment bonus problem) and the new locations are sited on the map shown in the HTML output file attached with this submission.

Conclusion

With a long-standing interest in this problem area, I was pleased to find the spatial data analytical tools right at my fingertips after taking this course. With a professional sponsorship, the dataset and accuracy could be increased to provide deeper proof of the ideal locations. Further research would also be required to determine land ownership, existing utilities, etc. to determine feasibility of the project sites. This is only the beginning of a fuller project planning effort. That is, the seeds of a great idea have been planted.